

MEMO.

WTW158 Calculus

Class Test 1 (Mo)

2 - 5 May 2023

Time: 15 min

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Surname:

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Initials:

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Student number:

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Use a pen to complete the test. The test is printed on both pages.

QUESTION 1 [1,1,2]

Find the derivative and do not simplify your answer.

(1.1) $g(t) = \tan(t) + \ln(t)$.

$$g'(t) = \sec^2 t + \frac{1}{t}$$

(1.2) $h(t) = e^t \cos(t^2)$

$$h'(t) = e^t \cos t^2 + e^t (-\sin t^2) \times 2t$$

(1.2) $f(x) = \frac{\sin(x)}{x^2 + 3x - 6}$

$$f'(x) = \frac{\cos x (x^2 + 3x - 6) - \sin x (2x + 3)}{(x^2 + 3x - 6)^2}$$

QUESTION 2 [2 marks]

Use the definition of a derivative to find the instantaneous rate of change of $x(t) = \frac{2}{t}$ when $t = 3$.

$$x'(3) = \lim_{h \rightarrow 0} \frac{x(3+h) - x(3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{2}{3+h} - \frac{2}{3}}{h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{2(3) - 2(3+h)}{3(3+h)} \right) \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{6 - 6 - 2h}{3(3+h)h} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{-2h}{3h(3+h)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{-2}{3(3+h)} = \frac{-2}{9}.$$

Instantaneous rate
of change at
 $t=3$ is $\frac{-2}{9}$.

QUESTION 3 [2 marks]

Use implicit differentiation to determine $\frac{dy}{dx}$ if: $\cos(x) + y^2 = 5y$.

$$\frac{d}{dx}(\cos x + y^2) = \frac{d}{dx}(5y)$$

$$\Rightarrow -\sin x + 2y \frac{dy}{dx} = 5 \frac{dy}{dx}$$

$$\Rightarrow 2y \frac{dy}{dx} - 5 \frac{dy}{dx} = \sin x$$

$$\Rightarrow \frac{dy}{dx}(2y - 5) = \sin x$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin x}{2y - 5}.$$

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QUESTION 1 [1,1,2]

Find the derivative and do not simplify your answer.

(1.1) $h(t) = \tan(t) + 3e^t$

$$h'(t) = \sec^2 t + 3e^t$$

(1.2) $f(t) = \sqrt{3t} - a^t = (3t)^{\frac{1}{2}} - a^t$

$$f'(t) = \frac{1}{2}(3t)^{-\frac{1}{2}} \times 3 - a^t \ln a.$$

(1.2) $f(x) = \frac{\sin(x)}{x^2 + 3x - 6}$

$$f'(x) = \frac{\cos x (x^2 + 3x - 6) - \sin x (2x + 3)}{(x^2 + 3x - 6)^2}$$

QUESTION 2 [2 marks]

Use the definition of the derivative to determine the derivative of $f(x) = \sqrt{x}$ in the point $x = 1$.

$$\begin{aligned}
 f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sqrt{1+h} - \sqrt{1}}{h} \quad \left(\frac{0}{0} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{\sqrt{1+h} - 1}{h} \right) \left(\frac{\sqrt{1+h} + 1}{\sqrt{1+h} + 1} \right) \\
 &= \lim_{h \rightarrow 0} \left(\frac{1+h - 1}{h(\sqrt{1+h} + 1)} \right) \\
 &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{1+h} + 1)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{1+h} + 1} = \frac{1}{1+1} = \frac{1}{2}
 \end{aligned}$$

QUESTION 3 [2 marks]

Use implicit differentiation to determine $\frac{dy}{dx}$ if: $\cos(y) + xy^2 = x$.

$$\begin{aligned}
 \frac{d}{dx}(\cos y + xy^2) &= \frac{d}{dx}(x) \\
 \Rightarrow -\sin y \cdot \frac{dy}{dx} + 1 \cdot y^2 + x(2y) \frac{dy}{dx} &= 1 \\
 \Rightarrow \frac{dy}{dx}(-\sin y + 2xy) &= 1 - y^2 \\
 \Rightarrow \frac{dy}{dx} &= \frac{1 - y^2}{-\sin y + 2xy}
 \end{aligned}$$

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Use a pen to complete the test. The test is printed on both pages.

QUESTION 1 [1,1,2]

Find the derivative and do not simplify your answer.

(1.1) $h(t) = 6 \sin(t) + at$

$$h'(t) = 6 \cos t + a$$

(1.2) $h(t) = \cos(t^2) + 2e^t$

$$h'(t) = -\sin t^2 \times (2t) + 2e^t$$

(1.2) $f(x) = \frac{\sqrt{x-2}}{e^x}$

$$f'(x) = \frac{\frac{d}{dx}(x-2)^{\frac{1}{2}} \cdot e^x - e^x \sqrt{x-2}}{(e^x)^2}$$

$$= \frac{\frac{1}{2}(x-2)^{-\frac{1}{2}} \cdot e^x - e^x \sqrt{x-2}}{e^{2x}}$$

QUESTION 2 [2 marks]

Find the equation of the line tangent to the graph of the function $y = e^x$ in the point $(2, e^2)$.

Eq. of tangent: $y = mx + c$

$$m = \left. \frac{dy}{dx} \right|_{x=2} = \left. \frac{d}{dx} e^x \right|_{x=2} = e^x \Big|_{x=2} = e^2$$

Eq. of tangent:

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - e^2 = e^2(x - 2) = e^2x - 2e^2$$

$$\begin{aligned} \Rightarrow y &= e^2x - 2e^2 + e^2 \\ &= e^2x - e^2 \end{aligned}$$

Eq. of tangent in the point $(2, e^2)$: $y = e^2x - e^2$

QUESTION 3 [2 marks]

Use implicit differentiation to determine $\frac{dy}{dx}$ if: $y^2 - 3x = 7y$.

$$\frac{d}{dx}(y^2 - 3x) = \frac{d}{dx}(7y)$$

$$2y \frac{dy}{dx} - 3 = 7 \frac{dy}{dx}$$

$$\Rightarrow 2y \frac{dy}{dx} - 7 \frac{dy}{dx} = 3$$

$$\Rightarrow \frac{dy}{dx}(2y - 7) = 3$$

$$\Rightarrow \underline{\underline{\frac{dy}{dx} = \frac{3}{2y - 7}}}$$

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QUESTION 1 [1,1,2]

Find the derivative and do not simplify your answer.

(1.1) $f(x) = 3 \cos(x) - x^3$

$$f'(x) = -3 \sin x - 3x^2$$

(1.2) $f(x) = a^{3x} \tan(x)$

$$f'(x) = (a^{3x} \ln a \times 3) \tan x + a^{3x} (\sec^2 x)$$

(1.2) $f(x) = \frac{\sqrt{x-2}}{e^x}$

$$f'(x) =$$

$$\frac{\frac{d}{dx}(x-2)^{\frac{1}{2}} \cdot e^x - \sqrt{x-2} e^x}{(e^x)^2}$$

$$= \frac{\frac{1}{2}(x-2)^{-\frac{1}{2}} \cdot e^x - e^x \sqrt{x-2}}{e^{2x}}$$

QUESTION 2 [2 marks]

Find the equation of the line tangent to the graph of the function $y = e^x$ in the point $(1, e)$.

$$y = \text{max}$$

$$m = \left. \frac{dy}{dx} \right|_{x=1} = e^x \Big|_{x=1} = e.$$

Equation of tangent line:

$$y - y_1 = m(x - x_1)$$

$$y - e = m(x - 1) \\ = e(x - 1)$$

$$\Rightarrow y = ex - e + e = ex.$$

Eq. of tangent in the point $(1, e)$ is $y = ex$.

QUESTION 3 [2 marks]

Use implicit differentiation to determine $\frac{dy}{dx}$ if: $x^2 + 2xy = 4y$.

$$\frac{d}{dx}(x^2 + 2xy) = \frac{d}{dx}(4y)$$

$$\Rightarrow 2x + 2y + 2x \frac{dy}{dx} = 4 \frac{dy}{dx}$$

$$\Rightarrow 2x \frac{dy}{dx} - 4 \frac{dy}{dx} = -2x - 2y$$

$$\Rightarrow \frac{dy}{dx}(2x - 4) = -2x - 2y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x - 2y}{2x - 4}.$$